

Jason Duran Dutton

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Professional Summary

Engineering professional with extensive experience maintaining, troubleshooting, and restoring complex electrical and mechanical infrastructure in safety-critical industrial environments. Proven record supporting large-scale underground operations through reliability-focused maintenance, emergency outage response, regulatory compliance, electrical systems oversight, fluid transport design, and operational risk management.

Experienced collaborating directly with engineers on pumping systems, infrastructure planning, controls integration, and production-critical problem solving where equipment performance directly affected personnel safety and operational continuity. Complements industrial engineering experience with independent research in scientific computing, mathematical modeling, quantum systems, computational physics, and simulation development.

Core Competencies

- Systems Engineering
 - Reliability Engineering
 - Root Cause Analysis
 - System Integration
 - Technical Risk Assessment
 - Failure Investigation
 - Technical Documentation
- Electrical Engineering
 - Industrial Electrical Systems
 - Power Distribution
 - Motor Controls
 - Pump Systems
 - Instrumentation
 - Preventive Maintenance
 - Electrical Safety
- Operations Engineering
 - Emergency Response
 - Production Recovery
 - Regulatory Compliance
 - Infrastructure Reliability
 - Safety-Critical Operations
 - Cross-Functional Coordination
- Software and Computing

- Python
- Git/GitHub
- Scientific Computing
- Modeling and Simulation
- Data Analysis
- LaTeX

Professional Experience

Underground Electrical Repairman / Outby Electrician

Alpha Metallurgical Resources / Contura

2022–2025

- Maintained operational readiness of mine-wide electrical and mechanical infrastructure supporting continuous underground operations.
- Conducted required inspections and weekly compliance documentation for all assigned beltlines, belt drives, and supporting infrastructure.
- Built and maintained a strong compliance record through disciplined inspection, preventative maintenance, and corrective-action management.
- Led troubleshooting and recovery efforts during production-critical failures affecting active mining sections.
- Coordinated personnel and technical resources to safely restore production following major equipment outages.
- Worked directly with mining engineers to design, size, install, and optimize underground pumping systems.
- Performed engineering calculations supporting water transport, pump performance, drainage requirements, and infrastructure reliability.
- Designed and implemented water-management systems balancing dust suppression, roadway safety, and beltline protection.
- Served as a primary technical resource during complex troubleshooting and emergency response activities.

Underground Electrician / Maintenance Foreman

Alpha Natural Resources

2010–2016

- Maintained and repaired electrical systems supporting continuous mining operations.
- Led maintenance activities across multiple operating areas.
- Performed troubleshooting of motors, controls, conveyors, pumps, and electrical distribution systems.
- Coordinated repairs while balancing safety, compliance, and production requirements.
- Supported reliability improvements through preventative maintenance and infrastructure upgrades.

Lead Electrician – Pilot Process Research Installation

Lone Mountain Refining / Virginia Tech Research Deployment

2015–2016

- Led electrical integration of a pilot-scale mineral processing facility.
- Coordinated controls, instrumentation, sensors, electrical installation, and commissioning activities.
- Supported validation and testing of advanced separation technology during extended operational campaigns.
- Worked alongside engineers, researchers, and technical stakeholders in a development environment.

Scientific Computing and Engineering Projects

Independent Engineering and Research Projects

- Developed Python-based scientific modeling frameworks and simulation tools.
- Created computational systems for numerical analysis, data visualization, and mathematical modeling.
- Built the SEFI-QEC Stack software environment for geometric quantum systems exploration.
- Authored technical manuscripts involving systems modeling, computational physics, differential geometry, and quantum information concepts.
- Applied engineering workflows to software development, technical documentation, simulation design, and research publication.

Education

Mountain Empire Community College

Electrical studies focusing on underground and surface mining systems

Brevard Community College / UCF

Coursework in Computer Information Systems and Business Management

Castlewood High School

High School Diploma